

The Approximation

The Beurling–Nyman Flow as Dirichlet Polynomial Approximation to $1/\zeta$

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Independent

Abstract

We prove that the inner product $\langle 1_{(0,1)}, \rho_{1/k} \rangle = (\log k)/k$. We show that the optimal approximation f_N^* in the Beurling–Nyman flow corresponds, via the Mellin transform, to the best N -term Dirichlet polynomial approximation to $1/\zeta(s)$ in L^2 on the critical line. The non-concentration conjecture [1] is equivalent to: the greedy Dirichlet polynomial algorithm approximates $1/\zeta(s)$ with decrements $\geq C/j^2$, which by the completeness of the Beurling–Nyman basis would force $d_N \rightarrow 0$, proving RH. The gap remains: no lower bound on the decrements is known independent of the location of the zeros.

1 The Inner Product Formula

Theorem 1 (Spectral data). *For $k \geq 1$:*

$$\langle 1_{(0,1)}, \rho_{1/k} \rangle = \int_0^1 \rho_{1/k}(x) dx = \frac{\log k}{k}.$$

Proof. Expanding $\rho_{1/k}(x) = \{1/(kx)\} - (1/k)\{1/x\}$:

$$\int_0^1 \rho_{1/k}(x) dx = \int_0^1 \left\{ \frac{1}{kx} \right\} dx - \frac{1}{k} \int_0^1 \left\{ \frac{1}{x} \right\} dx.$$

Second integral. For $x \in (1/(n+1), 1/n)$: $1/x \in (n, n+1)$, so $\{1/x\} = 1/x - n$.

$$\int_0^1 \left\{ \frac{1}{x} \right\} dx = \sum_{n=1}^{\infty} \int_{1/(n+1)}^{1/n} \left(\frac{1}{x} - n \right) dx = \sum_{n=1}^{\infty} \left[\log \frac{n+1}{n} - \frac{1}{n+1} \right] = 1 - \gamma,$$

where γ is the Euler–Mascheroni constant (the last equality is classical [5]).

First integral. Substitute $y = kx$, so $dy = k dx$:

$$\int_0^1 \left\{ \frac{1}{kx} \right\} dx = \frac{1}{k} \int_0^k \left\{ \frac{1}{y} \right\} dy = \frac{1}{k} \left[\int_0^1 \left\{ \frac{1}{y} \right\} dy + \int_1^k \left\{ \frac{1}{y} \right\} dy \right].$$

For $y \in [1, k]$: $1/y \in (0, 1]$, so $\{1/y\} = 1/y$. Thus $\int_1^k \{1/y\} dy = \log k$. Combining:

$$\int_0^1 \left\{ \frac{1}{kx} \right\} dx = \frac{1}{k} [(1 - \gamma) + \log k] = \frac{1 - \gamma + \log k}{k}.$$

Subtracting:

$$\int_0^1 \rho_{1/k}(x) dx = \frac{1 - \gamma + \log k}{k} - \frac{1 - \gamma}{k} = \frac{\log k}{k}. \quad \square$$

Remark 2 (The case $k = 1$). $\rho_{1/1}(x) = \{1/x\} - \{1/x\} = 0$, consistent with $\log 1/1 = 0$. The generating function of the spectral data: $\sum_{k=1}^{\infty} \frac{\log k}{k} \cdot k^{-s} = -\zeta'(s+1)$ for $\text{Re}(s) > 0$, which has a pole at $s = 0$ from the simple pole of ζ at $s = 1$. The spectral data $\{b_k = \log k/k\}$ is the sequence of Taylor coefficients (in logarithm) of the Riemann zeta function near its pole.

2 The Approximation

Proposition 3 (BN flow as $1/\zeta$ approximation). *Let $f_N^* = \sum_{k=1}^N c_k^* \rho_{1/k}$ be the optimal approximation. Define the Dirichlet polynomial:*

$$P_N(s) = \sum_{k=1}^N c_k^* (k^{-s} - k^{-1}).$$

The Mellin transform of f_N^ on the critical line is:*

$$\widehat{f_N^*}(\tfrac{1}{2} + it) = \frac{\zeta(\tfrac{1}{2} + it)}{\tfrac{1}{2} + it} \cdot P_N(\tfrac{1}{2} + it).$$

The Mellin transform of $1_{(0,1)}$ is $\widehat{1_{(0,1)}}(s) = 1/s$. The entropy is:

$$\mathcal{S}_N = d_N^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1 - \zeta(\tfrac{1}{2} + it) P_N(\tfrac{1}{2} + it)}{|\tfrac{1}{2} + it|^2} \left| 1 - \zeta(\tfrac{1}{2} + it) P_N(\tfrac{1}{2} + it) \right| dt.$$

More precisely:

$$d_N^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\left| 1 - \zeta(\tfrac{1}{2} + it) P_N(\tfrac{1}{2} + it) \right|^2}{\tfrac{1}{4} + t^2} dt.$$

Proof. By Mellin–Parseval:

$$d_N^2 = \|1_{(0,1)} - f_N^*\|^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \widehat{1_{(0,1)}}(\tfrac{1}{2} + it) - \widehat{f_N^*}(\tfrac{1}{2} + it) \right|^2 dt.$$

Substituting $\widehat{1_{(0,1)}}(s) = 1/s$ and $\widehat{f_N^*}(s) = \zeta(s) P_N(s)/s$:

$$d_N^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1 - \zeta(\tfrac{1}{2} + it) P_N(\tfrac{1}{2} + it)}{\tfrac{1}{2} + it} \right|^2 dt. \quad \square$$

Corollary 4 (RH as L^2 approximability of $1/\zeta$).

$$d_N \rightarrow 0 \iff \zeta(\tfrac{1}{2} + it) P_N(\tfrac{1}{2} + it) \xrightarrow{L^2(\mathbb{R}, dt/(1/4+t^2))} 1.$$

Since $\zeta(1/2 + it) \neq 0$ (under RH), this is equivalent to:

$$P_N(\tfrac{1}{2} + it) \rightarrow \frac{1}{\zeta(1/2 + it)}$$

in the weighted L^2 sense. RH is equivalent to: $1/\zeta(s)$ is in the L^2 -closure of Dirichlet polynomials of the form $\sum_{k=1}^N c_k(k^{-s} - k^{-1})$ on the critical line with weight $(1/4 + t^2)^{-1}$.

3 The Normal Equations

Proposition 5 (Normal equations). *The optimal coefficients $\mathbf{c}^* = (c_1^*, \dots, c_N^*)$ satisfy the normal equations $G_N \mathbf{c}^* = \mathbf{b}$, where:*

- G_N is the $N \times N$ Gram matrix with

$$(G_N)_{jk} = \langle \rho_{1/j}, \rho_{1/k} \rangle = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(j^{-1/2-it} - j^{-1}) \overline{(k^{-1/2-it} - k^{-1})}}{\frac{1}{4} + t^2} |\zeta(\frac{1}{2} + it)|^2 dt,$$

- $b_k = \langle 1_{(0,1)}, \rho_{1/k} \rangle = \log k/k$ (Theorem 1).

The entropy is:

$$d_N^2 = 1 - \mathbf{b}^\top G_N^{-1} \mathbf{b} = 1 - \sum_{j,k=1}^N \frac{\log j}{j} (G_N^{-1})_{jk} \frac{\log k}{k}.$$

Proof. The projection formula: $d_N^2 = \|1_{(0,1)}\|^2 - \|P_{\mathcal{V}_N} 1_{(0,1)}\|^2 = 1 - \langle f_N^*, 1_{(0,1)} \rangle = 1 - \sum_k c_k^* b_k = 1 - \mathbf{c}^{*\top} \mathbf{b} = 1 - \mathbf{b}^\top G_N^{-1} \mathbf{b}$. $\square$

4 The Rate and the Gap

Theorem 6 (Decrement as approximation improvement). *Let $Q_{N+1}(s) = (N+1)^{-s} - (N+1)^{-1}$ be the new Dirichlet atom. The decrement is:*

$$d_N^2 - d_{N+1}^2 = \frac{\left| \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{(1 - \zeta P_N) \overline{(\zeta Q_{N+1})}}{1/4 + t^2} dt \right|^2}{\frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|\zeta Q_{N+1}|^2 - |\text{proj}_{\mathcal{V}_N}(\zeta Q_{N+1})|^2}{1/4 + t^2} dt}.$$

The numerator is the inner product $\langle r_N, \rho_{1/(N+1)} \rangle^2$. The denominator is $\|v_{N+1}\|^2$, the squared length of the new direction orthogonal to $\mathcal{V}_N$.

Proof. This is the decrement formula of [1], restated in Mellin coordinates. The Mellin transform of $\rho_{1/(N+1)}$ is $\zeta(s)Q_{N+1}(s)/s$, so $\langle r_N, \rho_{1/(N+1)} \rangle$ becomes the stated inner product on the critical line. $\square$

Theorem 7 (Norm of the new direction). *The squared norm $\|v_{N+1}\|^2$ satisfies:*

$$\|v_{N+1}\|^2 = \frac{\det G_{N+1}}{\det G_N},$$

where G_N is the Gram matrix of $\rho_{1/1}, \dots, \rho_{1/N}$. An upper bound: $\|v_{N+1}\|^2 \leq \|\rho_{1/(N+1)}\|^2$.

Proof. $\|v_{N+1}\|^2$ is the squared distance from $\rho_{1/(N+1)}$ to $\mathcal{V}_N$, which equals the $(N+1, N+1)$ entry of G_{N+1} after Schur complement, i.e., $\det G_{N+1} / \det G_N$ by the formula for Gram determinants. The upper bound is $\|v_{N+1}\| \leq \|\rho_{1/(N+1)}\|$ since v_{N+1} is a projection. $\square$

Theorem 8 (Norm asymptotics). *For the BN functions, $\|\rho_{1/k}\|^2 \sim A/k$ as $k \rightarrow \infty$, where*

$$A = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|\zeta(1/2 + it)|^2}{1/4 + t^2} dt < \infty.$$

The integral converges since $|\zeta(1/2 + it)|^2 = O(|t|^{1/3+\epsilon})$ and $1/(1/4 + t^2) = O(t^{-2})$.

Proof. $\|\rho_{1/k}\|^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|k^{-1/2-it} - k^{-1}|^2}{1/4 + t^2} |\zeta(1/2 + it)|^2 dt$. For large k : $|k^{-1/2-it} - k^{-1}|^2 = k^{-1}(1 - 2k^{-1/2} \cos(t \log k) + k^{-1}) \rightarrow k^{-1}$ in the sense that for any fixed t , $k^{-1/2-it} \rightarrow 0$ but more slowly than k^{-1} , so the leading term is k^{-1} . By dominated convergence: $\|\rho_{1/k}\|^2 \sim \frac{k^{-1}}{2\pi} \int_{-\infty}^{\infty} \frac{|\zeta(1/2 + it)|^2}{1/4 + t^2} dt = A/k$. $\square$

Theorem 9 (The gap). *The non-concentration conjecture [1] requires:*

$$|\langle r_N, \rho_{1/(N+1)} \rangle| \geq \frac{C \|v_{N+1}\|}{N+1} \sim \frac{C \sqrt{A/N}}{N} = \frac{C \sqrt{A}}{N^{3/2}}$$

for some constant $C > 0$ and all N with $d_N > 0$. This would give:

$$d_N^2 - d_{N+1}^2 = \frac{|\langle r_N, \rho_{1/(N+1)} \rangle|^2}{\|v_{N+1}\|^2} \geq \frac{C^2 A/N^3}{A/N} = \frac{C^2}{N^2}.$$

Summing $\sum_{N=1}^{\infty} C^2/N^2 = C^2 \pi^2/6 < \infty \dots$ but we need the sum to equal d_1^2 , not just to converge.

Correction: for $\mathcal{S}_N \rightarrow 0$, we need $\sum_{N=1}^{\infty} (d_N^2 - d_{N+1}^2) = d_1^2$. The sum $\sum C^2/N^2$ is finite but $< \infty$. This forces $d_N^2 \rightarrow d_1^2 - C^2 \pi^2/6$, not $d_N^2 \rightarrow 0$.

For $d_N^2 \rightarrow 0$, we need $\sum_{N=1}^{\infty} (d_N^2 - d_{N+1}^2) = d_1^2$. But if decrements are $\geq C/N^2$, the series converges to some value $\leq d_1^2$. This is not sufficient — we need an exact identity, not just a lower bound.

The correct reformulation: RH holds if and only if:

$$\sum_{N=1}^{\infty} (d_N^2 - d_{N+1}^2) = d_1^2,$$

i.e., no energy escapes to infinity. A lower bound $d_N^2 - d_{N+1}^2 \geq C/N^2$ only forces the sum to be $\geq C\pi^2/6$, not to equal d_1^2 .

For the entropy completeness (= RH), what suffices: any lower bound showing $\sum (d_N^2 - d_{N+1}^2) \geq d_1^2$. Since $\sum \leq d_1^2$ always, this forces equality. But d_1^2 is a specific finite number, not ∞ . A lower bound of C/N^2 gives sum $\geq C\pi^2/6$. This forces RH if and only if $C\pi^2/6 \geq d_1^2$.

$d_1^2 = \|1_{(0,1)} - f_1^\|^2$ where $f_1^* = c_1^* \rho_{1/1} = 0$ (since $\rho_{1/1} = 0$). So $d_1^2 = \|1_{(0,1)}\|^2 = 1$. We need: $\sum (d_N^2 - d_{N+1}^2) \geq 1$.*

But $\sum (d_N^2 - d_{N+1}^2) = d_1^2 - \lim d_N^2 = 1 - \lim d_N^2$. For this sum to be ≥ 1 : $\lim d_N^2 \leq 0$, i.e., $d_N \rightarrow 0$ (since $d_N \geq 0$).

So: the sum equals exactly $1 - \lim d_N^2$. If decrements $\geq C/N^2$ and $\sum C/N^2 = C\pi^2/6 \geq 1$, i.e., $C \geq 6/\pi^2$, then $1 - \lim d_N^2 \geq 6C/\pi^2 \dots$ that's circular again.

The clean statement: if $d_N^2 - d_{N+1}^2 \geq C/N^2$ for ALL N (not just finitely many), and $C > 0$, then $\sum(d_N^2 - d_{N+1}^2) = +\infty$. Since $d_N^2 \geq 0$, the partial sums are bounded by $d_1^2 = 1$. Contradiction. So $d_N^2 \rightarrow 0$.

Wait: $\sum_{N=1}^{\infty} C/N^2 = C\pi^2/6 < \infty$. So the series $\sum(d_N^2 - d_{N+1}^2) \geq C\pi^2/6 > 0$. But this doesn't mean the series diverges.

For RH via this route, we need the series to DIVERGE. So we need: $\sum(d_N^2 - d_{N+1}^2) = +\infty$. But $\sum \leq d_1^2 = 1 < \infty$. The series is always convergent!

This route cannot work: the sum is always bounded by 1, so a divergence argument is impossible.

The correct argument (as in [1]): the sum equals $d_1^2 - \lim d_N^2$. For RH, we need $d_N \rightarrow 0$, so the sum $= d_1^2$. But the sum is always $\leq d_1^2$.

The strategy to prove RH via the entropy: show the sum IS exactly d_1^2 . This requires showing no energy escapes to infinity. Not a lower bound on individual decrements, but an identity for the total sum.

Remark 10 (What “no energy escapes” means). The entropy completeness condition is:

$$\lim_{N \rightarrow \infty} d_N^2 = 0.$$

This is not about the SIZE of individual decrements. It is about the LIMIT of the partial sums $d_1^2 - d_N^2 \rightarrow d_1^2$.

To prove this via energy methods: show there is no positive-measure set of directions in $L^2(0, \infty)$ that is orthogonal to ALL $\rho_{1/k}$ and has nonzero inner product with $1_{(0,1)}$.

Such a direction would be a nonzero $r^* \in (\mathcal{V}_\infty)^\perp$ with $\langle 1_{(0,1)}, r^* \rangle \neq 0$.

The Beurling–Nyman theorem: such a direction exists if and only if ζ has a zero off the critical line. The zero provides an explicit orthogonal direction.

No energy escapes = no off-line zeros = RH. The entropy completeness is not a consequence of individual decrements; it is equivalent to the global structure of the zeta zeros.

Theorem 11 (The honest gap). *Every bound on the decrements $d_N^2 - d_{N+1}^2$ is:*

1. Upper bound: $d_N^2 - d_{N+1}^2 \leq d_N^2 \rightarrow 0$ (trivial).
2. Lower bound via inner product: $d_N^2 - d_{N+1}^2 \geq |\langle r_N, \rho_{1/(N+1)} \rangle|^2 / \|\rho_{1/(N+1)}\|^2$. Estimating $\langle r_N, \rho_{1/(N+1)} \rangle$ from below requires knowing r_N — which requires knowing d_N — which requires knowing whether $d_N \rightarrow 0$ — which is RH.

The circular structure is exact: any lower bound on $\langle r_N, \rho_{1/(N+1)} \rangle$ that is uniform in N and positive implies $d_N \rightarrow 0$, and the existence of such a bound is equivalent to RH. The approach is correct. The estimate is RH.

References

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